

Project 1

Customer Churn Analysis Using Machine Learning

Objective

To predict whether a customer is likely to stop using a service (churn) based on behavioral, demographic, and service-related factors. This is a classic industry problem used in telecom, banking, SaaS, and e-commerce.

Project Description

In this project, learners will work with a structured customer dataset and analyze patterns that contribute to churn. This will involve data cleaning, EDA, feature engineering, and training a classification model.

The goal is to understand how companies retain customers by identifying high-risk segments.

Dataset Example (Telecom or SaaS Customer Data)

Columns may include:

- CustomerID
- Tenure
- MonthlyCharges
- TotalCharges
- Gender
- Age
- Contract Type (Monthly/Yearly)
- Payment Method

- Service Usage Stats
 - Churn (Yes/No – Target Variable)
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Expected Workflows

1. Data Preprocessing:

- Handle missing or incorrect values
- Convert categorical columns using encoding
- Scale numerical features if needed

2. EDA & Insights:

- Churn distribution analysis
- Contract type vs churn rate
- Correlation heatmaps
- Revenue impact of churn

3. Modeling:

- Build classification models: Logistic Regression / Decision Tree / Random Forest
- Evaluate using:
 - Accuracy
 - Precision & Recall
 - Confusion Matrix

4. Final Output:

- List top factors influencing churn

- Provide recommendations to reduce churn
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Deliverables

- Jupyter notebook with all code
 - Visualizations and insights section
 - Model performance report
 - 4-5 actionable business insights
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Skills Gained

- Classification modeling
- EDA
- Business analytics thinking
- Understanding customer risk patterns